

CS & IT ENGINEERING



Operating System

Basics of OS

One Shot



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Topics to be Covered



Topic

Operating System Definition

Topic

Types of Operating System

Topic

Dual Mode of Operation



Topic : Introduction

❑ **GATE Ranks:**

- 682 (2009) – 3rd year
- 19 (2010) – 4th year
- 119,440 etc.

❑ **Education:**

- ME from IISc Bangalore
- M. tech from BITS-pilani in Data Science

❑ **Work:**

- 17+ Year Teaching Experience
- 14+ in GATE/IES
- Worked in Cisco, Audience Communication



Topic : Operating System

- Software abstracting hardware
- Interface between user and hardware
- Set of utilities to simplify application development/execution
- Control program
- Acts like a government



Topic : OS Course Syllabus

6-8 marks weightage

Chapter Number	Chapter Name
1	Introduction ✓
2	Process Management ✓
3	CPU Scheduling ✓
4	Process Synchronization ✓
5	Deadlock
6	Memory Management & Virtual Memory ⇒ Imp.
7	File System
8	Disk Scheduling



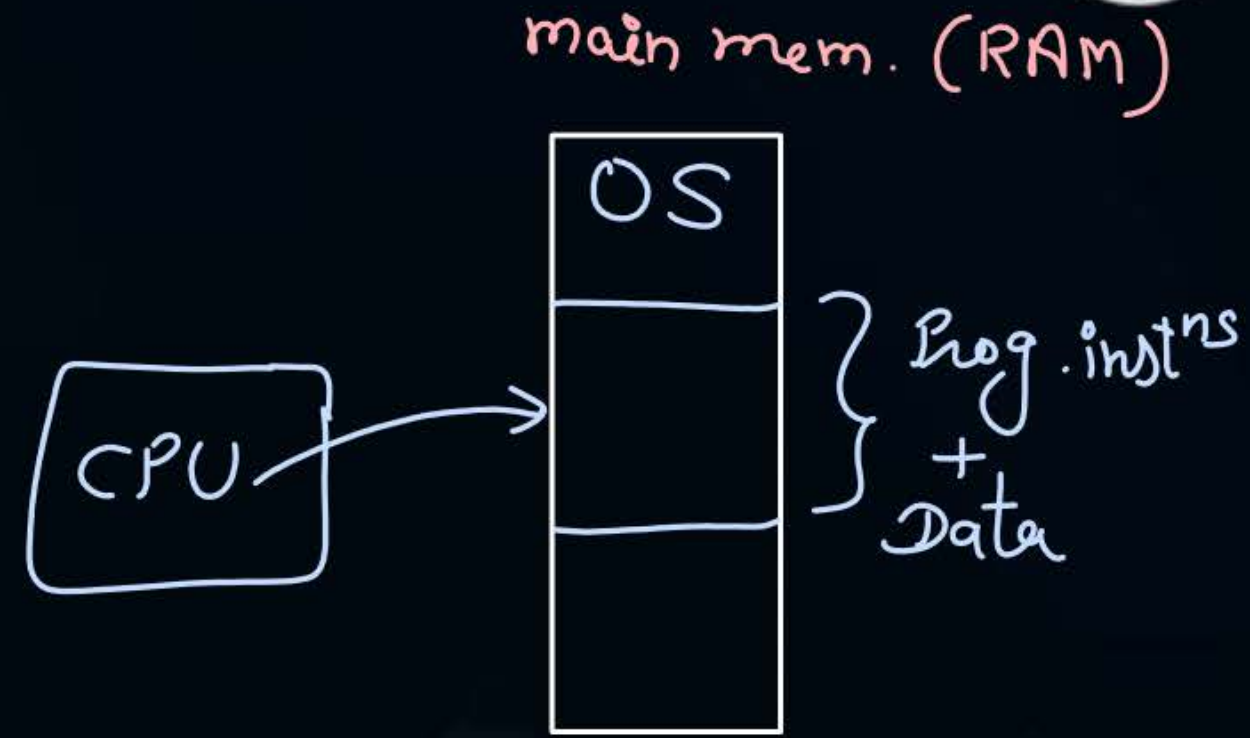
Topic : Services of OS



- User Interface
- Program Execution $\Rightarrow$ most imp. part
- I/O Operation
- File-System Manipulation

Communication (Inter-process Communication)

- Error Detection
- Resource Allocation
- Accounting
- Protection & Security





Topic : Types of OS



1. Uniprogramming OS
2. Multiprogramming OS
3. Multitasking OS (Time Sharing)
4. Multiprocessing OS
5. Multiuser OS
6. Real Time OS
7. Embedded OS
8. Handheld Device OS



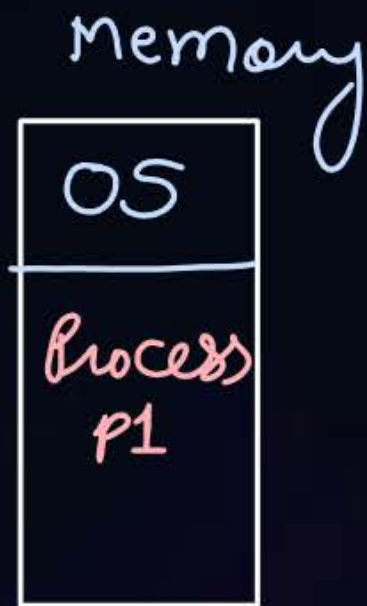
Topic : Uniprogramming OS

only one program at a time can reside in main memory.

If the single process goes for I/O operation
then CPU becomes idle.



CPU utilization is low





Topic : Multiprogramming OS

It allows more than one program to be in main memory (m.m.) at a time. If one program goes for I/O operatⁿ then another program can occupy CPU.

↓
better CPU utilization than uniprogramming

memory

OS
P1
P2
P3
P4

Degree of multiprogramming :-

no. of processes in main memory at a time.

multiprogramming as

```
graph TD; A[multiprogramming as] --> B[Non-preemptive]; A --> C[Preemptive];
```

Non-preemptive

A running process can leave CPU by its own.

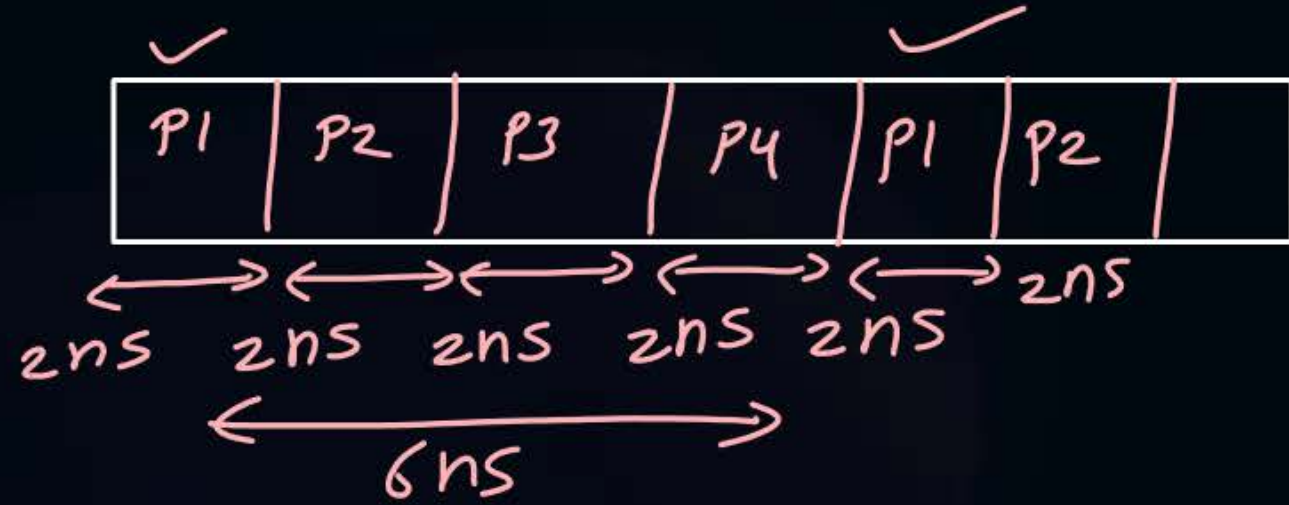
Preemptive

A running process can be taken out of the CPU forcefully.



Topic : Multitasking OS

It is an extension of multiprogramming OS in which processes are executed in round-robin manner.



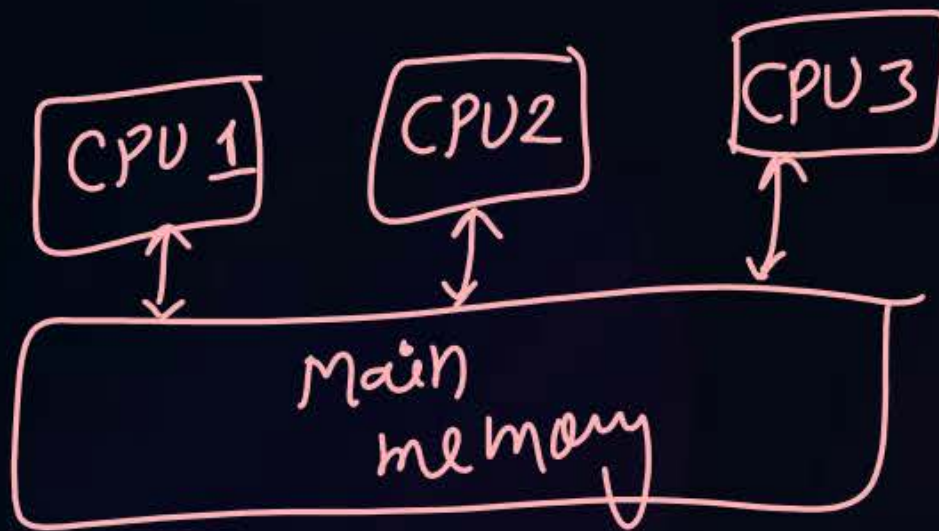


Topic : Multiprocessing OS

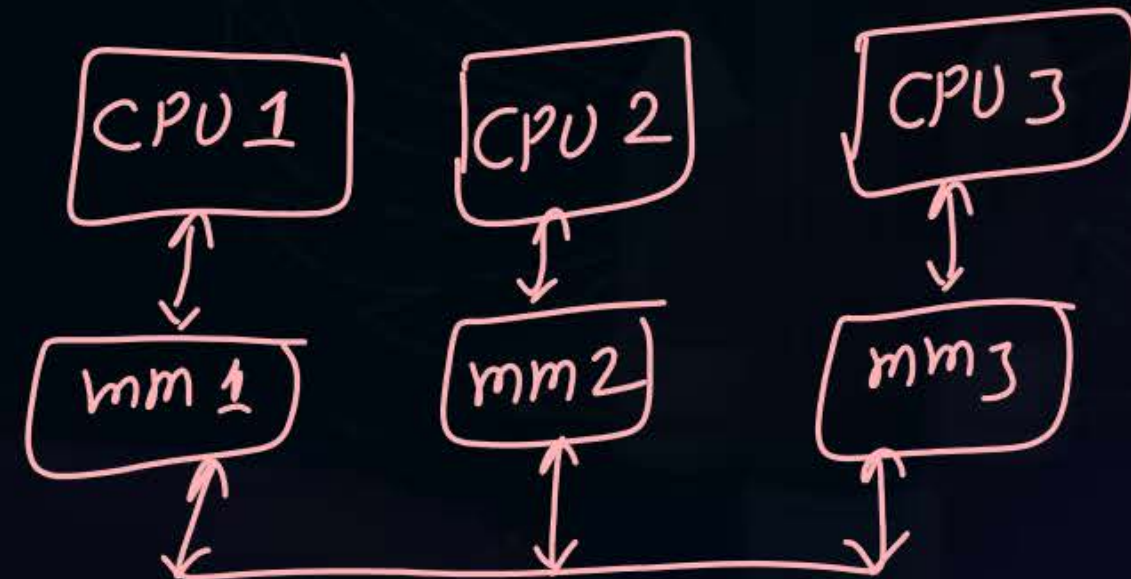
This OS works on computer system with multiple CPUs.

Types

Tightly coupled
(Shared memory)



Loosely coupled
(Distributed)

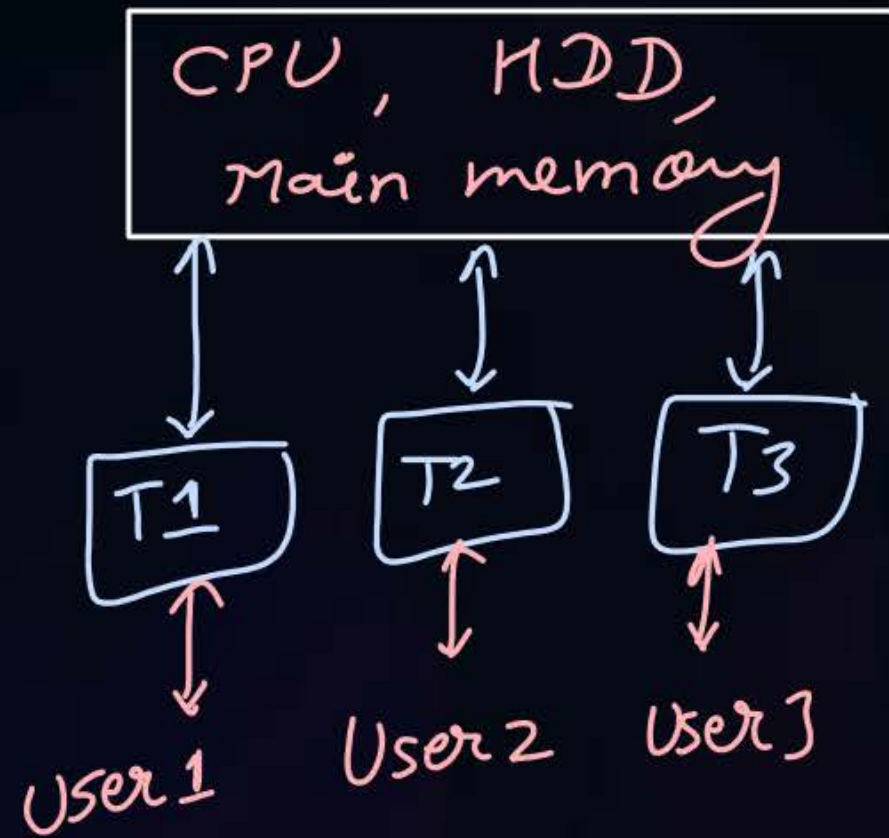




Topic : Multiuser OS

An OS which allows multiple users to use one computer system simultaneously.

Apple created OS } $\Rightarrow$ not multiuser
Microsoft - 11 - }



T1, T2, T3 $\Rightarrow$ Terminals
(keyboard, mouse, screen etc.)



Topic : Real Time OS



os works on real time Data or event.

os assigns deadline to each process.





Topic : Embedded OS



↓
OS used on embedded systems.

User interaction to OS is minimum



Topic : Handheld Device OS



os working on handheld devices



2 mins Summary

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Operating System Definition

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Types of Operating System

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Happy Learning

THANK - YOU